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A comprehensive analysis of repurposing abandoned oil wells for different energy uses: Exploration, applications, and repurposing challenges

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ABSTRACT

The rise in abandoned oil wells across the globe poses a serious environmental and public health risk. These wells, which are frequently abandoned by defunct corporations or owing to regulatory gaps, pose substantial risks. They have the potential to leak methane, a potent greenhouse gas, and contaminate groundwater. Researchers estimate that there are between 2 and 3 million abandoned oil and gas wells in the United States. Out of these, over 117,000 wells, spread across 27 states, are classified as "orphaned", and lack an identifiable party responsible for managing leakage or pollution risks. The escalating number of abandoned oil wells in the United States presents a dual challenge and opportunity in the realm of renewable energy. The global utilization of geothermal energy is on the rise, with approximately 72 countries harnessing this resource for various applications. About 24 of these countries generate electricity using geothermal energy through binary or flash cycle methods. The United States leads in geothermal electricity production, generating approximately 17,917 GWh annually. Global raise in geothermal energy utilization provides presents an opportunity to repurpose abandoned oil wells for geothermal energy production especially in the United States. These wells, often still possessing high temperatures and temperature gradients, can be converted into valuable geothermal resources, thus providing a sustainable energy solution and addressing the environmental hazards posed by the abandoned wells. This paper critically examines the feasibility of repurposing these wells for geothermal energy production, a strategy that offers a promising solution to both environmental hazards and the need for sustainable energy sources. Focusing on the technical, economic, and social dimensions, we present a comprehensive analysis that includes a case study of the Williston Basin in North Dakota, highlighting its potential for geothermal exploitation. Our approach employs Fourier's law of conduction to estimate the temperature at the bottom of selected wells. We address the critical challenges in this endeavor, ranging from ensuring the mechanical integrity of aging wells to navigating the economic and social implications of their repurposing. Our findings suggest that while significant challenges exist, especially in retrofitting old wells for new uses and garnering stakeholder consensus, the conversion of abandoned oil wells into geothermal energy sources is a viable and environmentally beneficial path forward. Finally representing a detailed exploration of their various potential geothermal and various applications This research contributes to the growing body of literature on sustainable energy solutions, offering practical insights and guidelines for future field implementations in the transition from fossil fuels to renewable energy sources.

1. Introduction

Geothermal energy is the heat stored beneath the earth's surface. Since its formation, Planet Earth has cooled down. The planet's formation around 4.5 billion years ago and the radioactive decay of naturally occurring isotopes primarily cause heat formation (Glassley, 2014). Manzella, (2019) suggests that heat travels from the earth's interior to the surface. The earth's heat flow is considered to be 57 mW/m2 in the continental crust and 99 mW/m2 in the oceanic crust (Barbier, 2002). The heat decreases from the earth's interior to the surface, and vice versa, from the exterior to the interior. We refer to this phenomenon as the geothermal gradient, a variation of heat with depth, with the highest values observed in volcanic areas where magma is near the surface (Manzella, 2019; Arndt, 2015). Different types of applications utilize geothermal energy, including direct heating, direct electricity generation, and indirect use for geothermal heat pumps (CSUE, 2015). In the

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United States, the direct utilization of geothermal energy includes pool heating, agricultural drying, industrial applications, grid applications, and ground-source heat pumps. The number of heat pumps is higher in the Midwest and mid-Atlantic regions, from North Dakota to Florida (Lund et al., 2005). Normal temperatures for geothermal electricity production range from about 180 °C to lower levels. Around 72 countries utilize geothermal energy, while about 24 countries produce electricity using the binary or flash cycle method (Bertani, 2007). El Salvador, Costa Rica, and Nicaragua are among the top six countries for electricity production from geothermal sources in Central America. The USA generates approximately 17,917 GWh annually, making it the top producer (Bertani; Lund et al., 2005).

Generally, these geothermal plants inject water to absorb heat from the subsurface, and then use this hot water to convert the refrigerants in the power cycles into vapor, which in turn powers the turbine generators for electricity production. Systems of this kind come with certain problems that are difficult to solve, like wastewater, drilling costs, scaling, and corrosion problems (Stáhl et al., 2000; Xianbiao et al., 2011). The drilling cost makes up 50% of a geothermal plant's total commissioning cost.

In the USA, crude oil production will be around 11,600,000 barrels per day in 2022, which is less than the previous year. The increase in oil well abandonment and economic feasibility issues is a significant reason for this decline (EIA; Nian and Cheng, 2018). Due to the COVID-19 pandemic, oil firms are less financially able to invest in well plugs and find it difficult to get new well bonds (DMR, 2020). The total number of documented orphaned wells in the United States was 81,857 from June to September 2021 and 123,318 from January to April 2022, representing 2% and 3% of all the estimated abandoned wells in the USA (Williams et al., 2021). Using these abandoned wells for geothermal energy production would not only reduce drilling costs but also address scaling, corrosion, and reinjection issues (Xianbiao et al., 2011).

Every well experience a plateau in its oil production rate several years after the initial production and buildup phase. A reduction in geothermal formations' physical properties and petroleum fluid characteristics causes the well to enter a decline phase (Sorrell et al., 2012). During the decline phase, the fields will lose their ability to meet the rate of demand, and at this point, they will reach their economic limit. Companies consider oil wells abandoned when the revenue from production fluids falls below the set cost of operations (Wang et al., 2016; Höök et al., 2014). Companies can retrofit existing abandoned wells into geothermal wells by sealing the insulation bottom, resulting in a double pipe configuration for both horizontal and vertical oil wells, according to Cui et al., 2017. Thermal energy production depends on the fluid flow rate and geothermal gradient, which decrease with the age of the wells (Kujawa et al., 2006).

Recently, many researchers have paid attention to abandoned oil wells. For instance, Kujawa et al. (2006) studied the utilization of deep geothermal wells and inferred that the flow rate and insulation have important effects on heat transfer. Davis and Michaelides (2009) studied geothermal power production from abandoned oil wells, taking into account local geothermal gradients and well depths. However, their research assumed the rocks' temperature to remain constant over time, suggesting that the calculated power generation in their study exceeded the actual value.

In recent years, efforts to repurpose abandoned oil wells into geothermal resources have increased. However, there are many principal challenges that restrict further development in this particular area (Santos et al., 2022). Common challenges include low energy conversion efficiency, incorrect planning and evaluation, regulatory and legal issues, and well integrity difficulties. The well selection for repurposing into geothermal wells depends on the geothermal gradient of the well and the proximity of the end users (Santos et al., 2022; Wang et al., 2018).

When converting oil and gas wells into geothermal wells, the process of characterizing the reservoir, conducting geomechanical modelling, and determining optimal methods for extracting maximum oil and gas from the ground provides valuable data for subsurface studies. It is noteworthy that these studies require intensive lab testing and data acquisition—such as seismic data—which are highly capital-intensive yet remain free or inexpensive for future repurposing designs. However, such collections of information might not be accessible for many older wells (Santos et al., 2022). Currently, a lack of mature utilization technologies, low resource exploration levels, and overall policies, incentives, and standards limit geothermal energy exploration in oil fields. These and other important factors make it challenging to develop large-scale geothermal utilization technology (Wang et al., 2016).

The transition from fossil fuels to alternative energy sources is a complex and multifaceted challenge that requires careful consideration of technical, economic, social, and environmental factors (Townsend-Small et al., 2016). Proposals have suggested repurposing existing oil and gas wells for alternative energy and utilization techniques like geothermal energy, hydrogen storage, and carbon capture, utilization, and storage (CCUS) as a potential solution (Cano et al., 2022; Mehmood et al., 2019; Josiane et al., 2022).

The idea of repurposing existing wells is appealing due to the already established infrastructure for drilling and extraction (Watson et al., 2020; Caulk and Tomac, 2017). However, this endeavour presents significant challenges that require attention (Lebel et al., 2020; Liu et al., 2018; Soldo et al., 2020).

One of the key challenges is ensuring the mechanical integrity of the wells. Abandoned oil and gas wells can pose risks of methane emissions and environmental contamination if not properly sealed and monitored (Lebel et al., 2020). It is crucial to assess the integrity of these wells and develop effective methods for repurposing them without compromising safety or environmental protection (Lebel et al., 2020). Economic considerations also have a significant impact on the feasibility of repurposing wells. We must carefully evaluate the cost of retrofitting and repurposing wells for alternative energy systems, taking into account factors like potential energy output, market demand, and regulatory requirements (Watson et al., 2020; Caulk and Tomac, 2017). Additionally, the economic viability of repurposing wells may vary depending on the specific location and geological characteristics (Watson et al., 2020).

Social and environmental considerations are equally important in the repurposing of oil and gas wells. Stakeholder engagement and community acceptance are crucial for the successful implementation of alternative energy projects (Thomas et al., 2021). Furthermore, we must carefully assess the environmental impact of repurposing wells to ensure that the benefits of alternative energy production outweigh any potential negative consequences (Townsend-Small et al., 2016).

In the subsequent sections, we will delve into the specifics of these challenges, offering a comprehensive understanding of the complexities involved in repurposing oil and gas wells for alternative energy applications. We aim to shed light on the technical, economic, social, and environmental issues that require attention through this discussion, providing a comprehensive perspective on the matter.

Exploiting geothermal energy from abandoned wells is currently considered a research hotspot internationally. Pilot operations and feasibility studies conducted around the world indicate the validity of this concept, albeit at a low efficiency, opening the door for potential advances in this field (Santos et al., 2022). Globally, researchers have identified several million abandoned or orphaned oil and gas exploratory wells. In the USA, there are about 310,000 to 800,000 or more abandoned oil wells in the country (Liu et al., 2023). The Environmental Protection Agency (EPA) of the United states of America (EPA, 2017) reports that nearly 3.2 million abandoned wells remain unutilized for production, injection, or any other purposes. Unplugged, abandoned wells make up about 2.1 million of these 3.2 million wells, according to estimates. Fig. 1 shows the estimated number of documented and undocumented idle or orphaned wells in the United States (Raimi, 2019).

As a result, this paper focuses on investigating the potential of geothermal energy extraction and alternative energy sources from

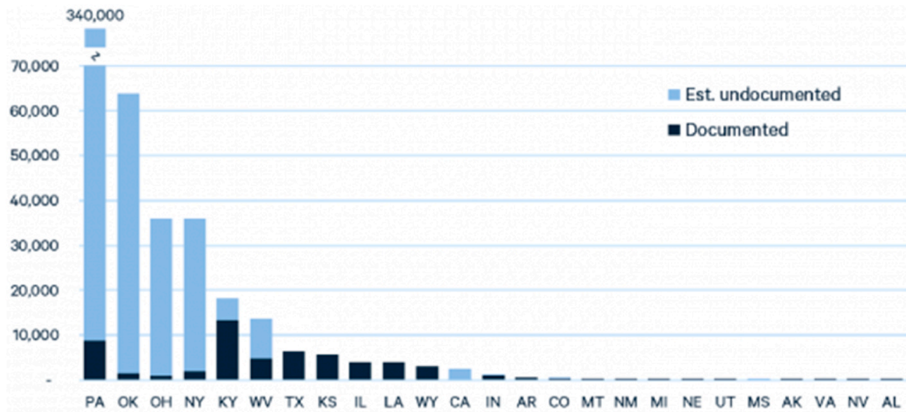

Fig. 1. Estimated undocumented and Documented wells in us (Raimi, 2019).

abandoned oil wells. The paper assesses the temperature at the bottom of these abandoned wells to pinpoint those with potential for geothermal energy use, categorizes classic cases of geothermal energy exploitation from these wells, and ultimately underscores the technological challenges and significant issues that require future resolution. This paper aims to serve as reference material and a viable foundation for the repurposing aspects of abandoned oil and gas wells.

2. Case study on temperature availability in oil wells at specific depths situated in North Dakota

The geothermal gradient is the increase in temperature that occurs for each unit of depth within the Earth. Despite regional variations, this gradient typically ranges from 25 to 30 °C/km, or 15 °F/1000 feet (van der Meer et al., 2014). This gradient can be very steep in volcanic regions. Understanding the regional geothermal gradient is essential for engineers working with fluids in deep wells. For this research, we analyse the Red River formation in North Dakota, within the Williston Basin.

The Williston Basin is a large intracratonic sedimentary basin located in southern Saskatchewan (Canada), western North Dakota, and eastern Montana illustrated in Fig. 2. The basin is rounder and deeper as it gets closer to the center. Its sedimentary strata, which can be several kilometres deep, provide evidence of a lengthy history of environments associated with the sea, lakes, and rivers (Carlson and Anderson, 1965). The Williston Basin is an extensive epicratonic basin located beneath the Great Plains, encompassing North Dakota, South Dakota, Montana,

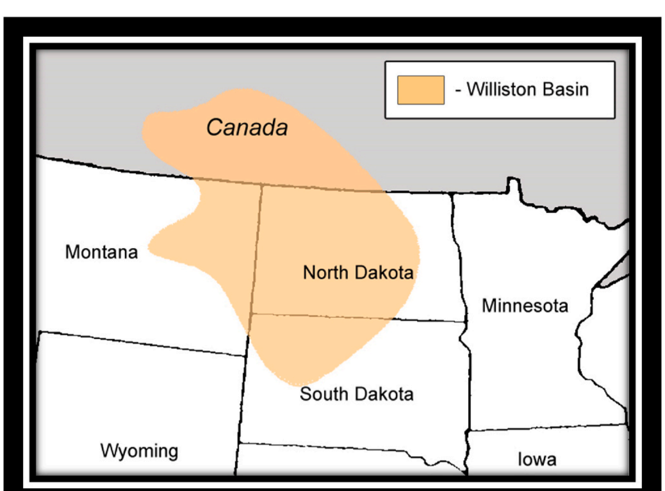

Fig. 2. Distribution of Williston Basin across states (Williams et al., 2021).

Saskatchewan, and Manitoba. The Sioux Arch, extending into the Dakotas and southeastern Manitoba, borders it to the east, the Meadow Lake Escarpment from the lower Paleozoic epoch to the north, and the Sweetgrass Arch in northern Montana and southern Alberta to the west (Husinec, 2016).

The Williston Basin (Fig. 3) in North Dakota is a sedimentary basin underlain by Precambrian crystalline igneous and metamorphic rocks. Moving upward, the Cambrian period includes the Deadwood Formation, which is composed of sandstones and shales, followed by Ordovician carbonates such as the Red River Formation. Carbonates and evaporites from the Silurian and Devonian periods are also present, particularly in the Duperow and Birdbear formations. Carbonates and shales, such as the Madison Group, characterize the Mississippian and Pennsylvanian periods. The Permian period brings red beds and evaporites, followed by Triassic red beds and Jurassic marine shales and sandstones, such as the Piper and Rierdon formations. The Cretaceous epoch is notable for its vast marine deposits and the hydrocarbon-rich Bakken Formation. The Tertiary period, which includes the Fort Union Formation, delivers non-marine sediments, while the Quaternary period concludes with glacial deposits (Gaswirth et al., 2010).

The Red River Formation is divided into upper and lower subunits. The upper Red River Formation has three brining cycles, defined as A, B, and C cycles. Nodular to laminated anhydrite, burrowed lime mudstone to fossil wackestone, laminated microcrystalline dolostone, and thin argillaceous dolomitic mudstone are all parts of the upper Red River Formation (Longman et al., 1983; Fox, 1993). The burrow-mottled lime mudstone and fossil wackestone, which have undergone varying degrees of dolomitization towards the top of the interval, compose the lower Red River. Hydrocarbon reservoirs with 10–20% make up the “D" zone porosity. The “D" zone consists of dolomitized, burrow-mottled facies. Additionally, the potential petroleum source beds are located in the “D" zone (Kendall, 1976). The formation has varied porosity, with major porosity zones designated as “A,” “B,” and “C" anhydrite. The presence of evaporites and dolomitization influences the porosity in these zones. Different parts of the rock have different levels of permeability, and the levels are usually higher in dolomitized areas where secondary pores form during the dissolution and recrystallization processes (Tanguay and Friedman, 2001).

For this research, we are identifying oil wells with production pools in the Red River Formation found in McKenzie County, North Dakota. Gosnold et al. (2012) found that the Red River Formation in specific western counties in North Dakota (McKenzie, Williams County, etc.) has a higher temperature and temperature gradient, around 140–160 °C shown in Fig. 4, and is almost 230 m thick. The Red River Formation is deepest in the western part of North Dakota, especially in McKenzie County, where it is approximately 12,000 feet deep. The formation is widespread throughout western North Dakota, particularly west of the Nesson anticline, where structural closures play an important role in

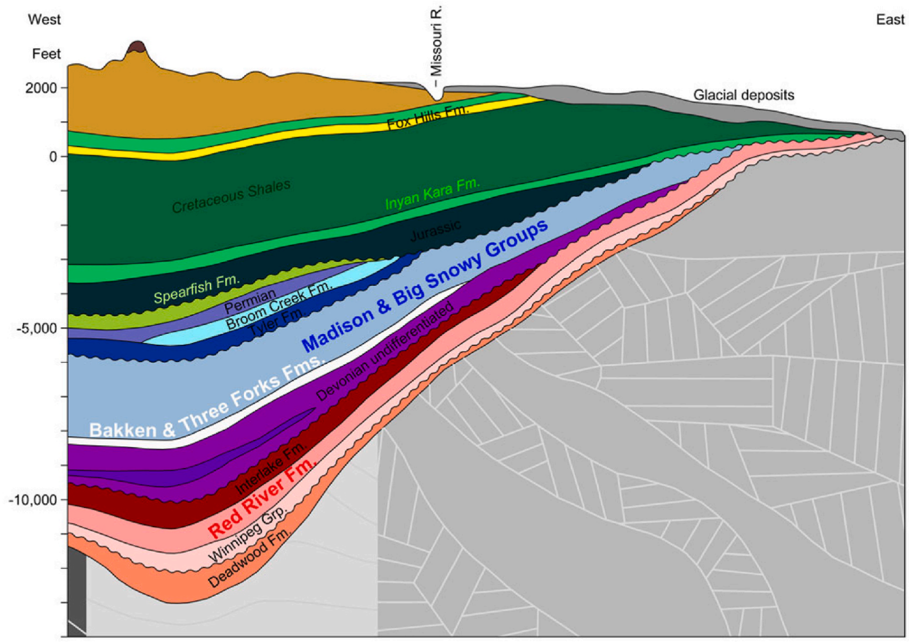

Fig. 3. Cross section of Williston basin, North Dakota (North Dakota Geological Survey, 2015).

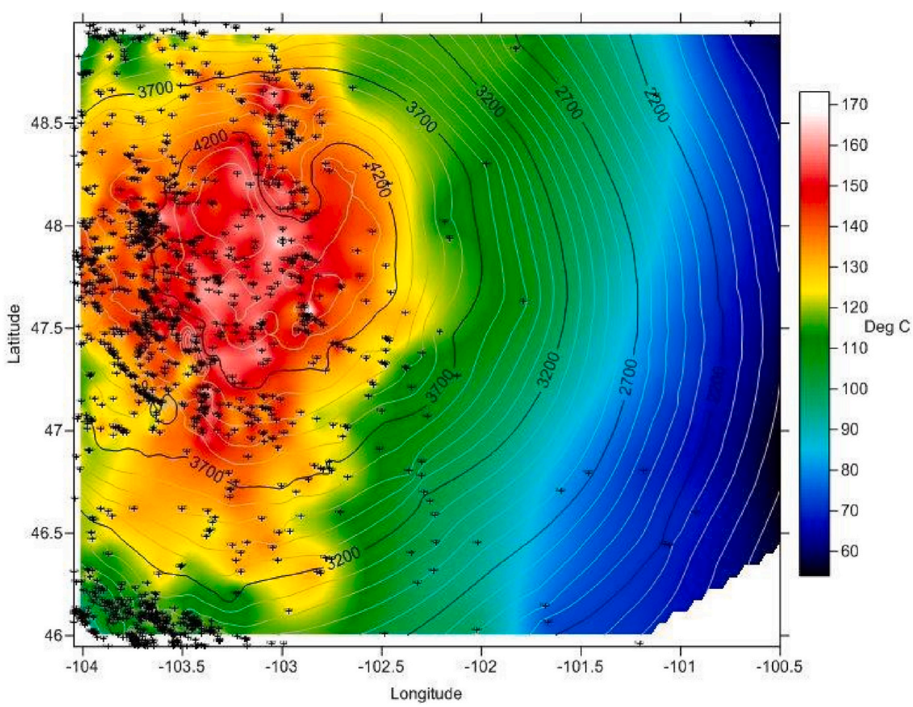

Fig. 4. Temperature map of Red River formation in Western North Dakota (Gosnold et al., 2012). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1
Abandoned oil wells which have the production pool of Red River formation in McKenzie County, Williston basin, North Dakota (Department of mineral resources, 2022).

File No	County Name	Field Name	Produced Pools	Wellbore	Latitude	Longitude	Well Type	Well Status	Depth (Feet)
5163	Bowman	Coyote Creek	Red River	Vertical	46.164408	−103.691425	OG	AB	9696
15,569	Mckenzie	Moline	Red River	Vertical	47.683775	−103.794864	OG	AB	13,286
15,670	Mckenzie	Alexander	Red River	Vertical	47.88089	−103.569356	OG	AB	13,505
34,442	Mckenzie	Lonesome	Red River	Directional	47.901901	−103.678514	OG	AB	13,302

hydrocarbon accumulation (Anna, 2013). In terms of hydrocarbon output in North Dakota, the Red River Formation is second only to the Bakken Formation.

The data for this research is taken from the scout ticket data of the North Dakota Oil and Gas division (NDGS) database represented in Table 1. The calculation performed gives the behavior of the temperature with respect to the depths of specific abandoned wells. This calculation can be used to estimate the geothermal potential of the abandoned wells.

The temperature estimation for each oil wells is performed by using the concept of Fourier law of conduction which express the relationship between heat flow, temperature gradient and thermal conductivity. Heat flow is the product of temperature Gradient and thermal conductivity.

$$Q = \lambda\!\!\!{}^{-} \frac{\Delta t}{\Delta z} \tag{1}$$

$$\Delta t = Q \frac{\Delta z}{\lambda\!\!\!{}^{-}} \tag{2}$$

Equation (2) can be represented for a depth n (tn) calculated by adding the temperature change associated with each stratigraphic unit

$$t_n = t_0 + Q\left(\frac{\Delta z_1}{\lambda_1} + \frac{\Delta z_2}{\lambda_2} \dots\dots\dots + \frac{\Delta z_n}{\lambda_n}\right) \tag{3}$$

- t = Temperature
- λ = Thermal conductivity
- $\frac{\Delta t}{\Delta z}$ = Temperature Gradient
- Q = Heat Flow
- n = Number of overlying stratigraphic units
- λ_n = Conductivity of the nth layer
- z_n = Depth of nth layer
- t_0 = Average surface temperature
- t_n = Temperature of nth unit.

To establish the temperature at a certain place, one needs first to know the average surface temperature, the thickness of the layers as determined by well logs, and the thermal conductivities of the geological strata taken from the literature.

The reference data for this calculation assumes a heat flow (Q) of 0.054 W/m2, an annual mean temperature of 6 °C, and a thickness of the first formation (Green Horn) of around 155 m (McDonald, 2015). These values are based on data from the Department of Mineral Resources, North Dakota, which provides specific geological information about the Williston Basin. From the point of view of McKenzie County, the temperature, thickness, and heat flow of the formations are reasonable for implementing enhanced geothermal systems, according to McKenzie County. Table 2 provides the thermal conductivity of the formations from top to bottom until the Red River Formation, which is essential for estimating the temperature at different depths.

2.1. Results

In the Williston Basin, known for its varied geological formations, the analysis of temperature and gradient data from four distinct wells reveals a promising scenario for geothermal energy applications. Each well displays a consistent increase in temperature with depth, a key indicator of geothermal potential. Notably, Wells 2 and 4 represented in Figs. 6 and 8 stand out with steeper temperature gradients, suggesting a higher heat flow that is favorable for electricity generation. In contrast, Wells 1 and 3, while exhibiting gentler gradients, achieve temperatures conducive to direct-use applications like heating represented in Figs. 5 and 7. This variance in temperature gradients underscores the significance of depth in determining the most appropriate geothermal technology for each well.

Table 2
Thermal conductivity of the formations from top to bottom, Williston Basin (DMR, 2015).

Formation	Thermal conductivity (W/m/K)
Green Horn	1.1
Mowry	1.1
Iniyan Kara	1.4
Swift	1.2
Rierdon	1.6
Spear Fish	3
Minnekatha	4
Opecha	4
Eroded Bottom Creek Amsden	4
Tyler	1.2
Big Snowy	2.4
Kimber Lime	2.7
Madison Group	2.5
Mississippian Ratcliff	2.5
Mississippian Base of last salt	2.5
Mississippian Frobisher Alida	2.5
Mississippian Lodgepole	2.5
Mississippian/Devonian Bakken	1.1
Three Forks	1.3
Bird Bear	3.1
Duperow	1.8
Souris River	2.6
Dawson Bay	2.8
Prairie	2.8
Winnipegosis	2.8
Inter Lake	2.8
Cedar Lake	2.8
Gunton	2.8
Stoughton	2.8
Red River	3.8

The cumulative data from these wells paints a picture of a geologically diverse subsurface, with each segment having its own unique thermal characteristics. This diversity offers the opportunity to tailor geothermal solutions to specific conditions, ranging from direct heating systems to more complex electricity generation setups. The findings, however, warrant further investigation. Detailed geophysical and geological studies are essential to confirm these preliminary observations and accurately delineate the full scope of the geothermal resources available in the Williston Basin. These studies will be instrumental in guiding the strategic development of geothermal energy in the region and optimizing the use of its subsurface thermal resources.

Obtaining geothermal energy from oil wells also requires dealing with obstacles such as maintaining the well's integrity, managing thermal stress, and ensuring chemical compatibility. The wellbore, casing, and cementing materials may be unable to endure elevated temperatures and caustic geothermal fluids. The reservoir must also possess appropriate parameters, such as temperature, pressure, and fluid flow (Kurnia et al., 2021). Specialized heat exchangers and pumps are necessary for technological adjustments.

3. Challenges involved in repurposing oil wells for geothermal use

3.1. Well integrity challenges

As the energy landscape shifts from traditional fossil fuel extraction to alternative applications, wells, both old and new, find themselves at the forefront of these transformations. Historically, well construction in the energy sector was influenced by the best practices and materials available at the time (McDaniel et al., 2014). While they served their initial purpose, many of these legacy wells lack the technological sophistication and resilience found in modern engineering. This distinction between the old and new approaches becomes especially relevant as wells transition into novel operational paradigms (Merzoug and Okoroafor, 2023).

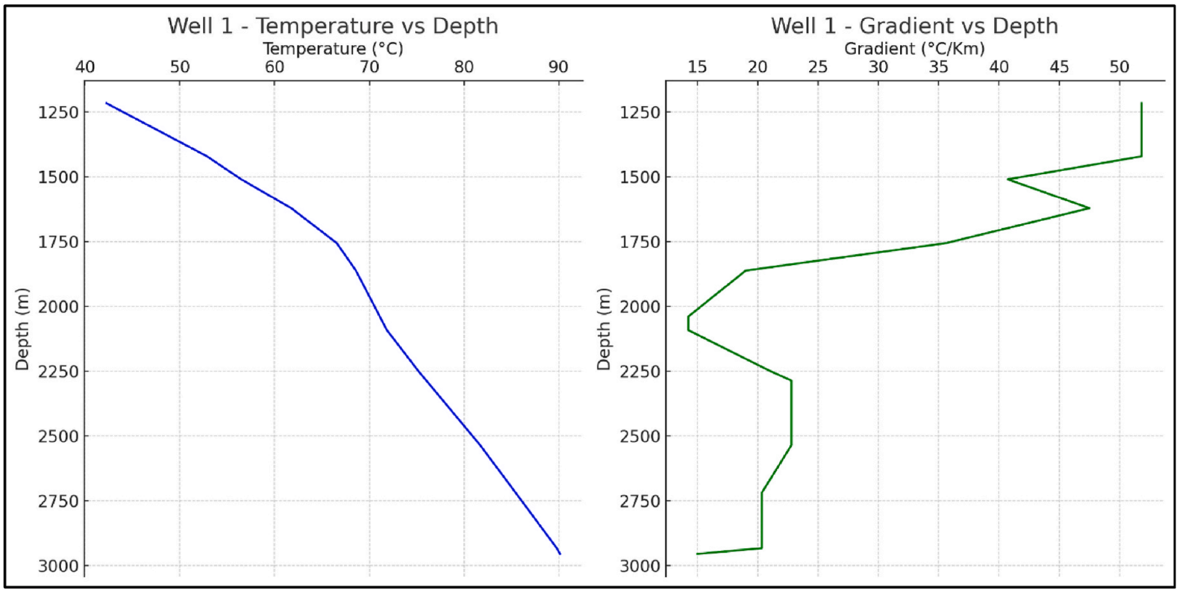

Fig. 5. Behavior of temperature and gradient of the well with respect to depth.

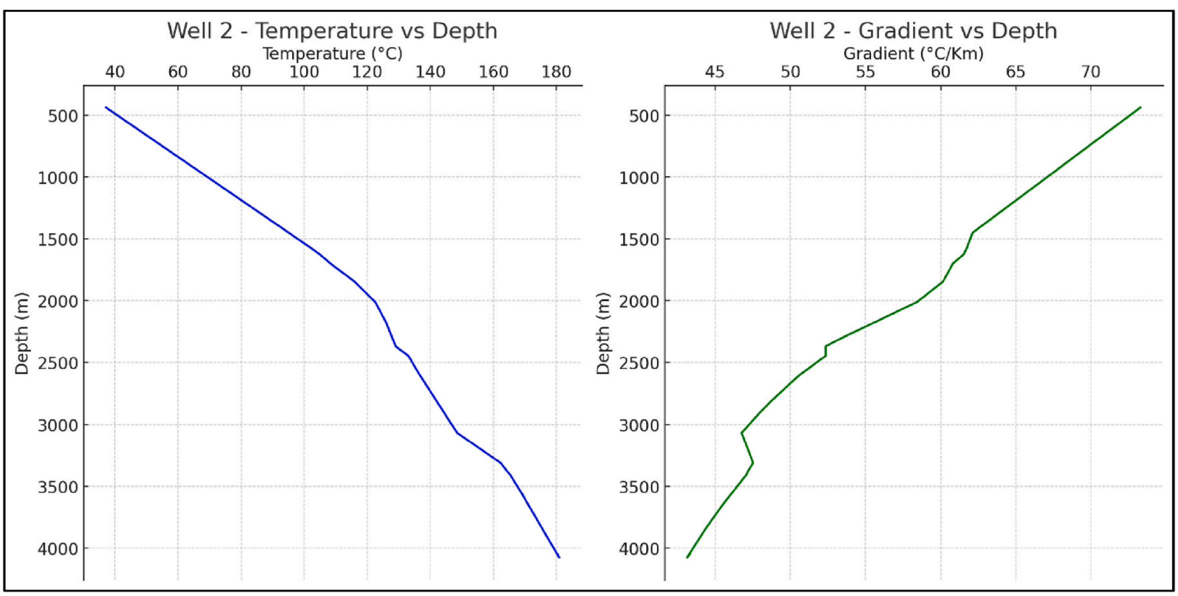

Fig. 6. Behavior of Temperature and gradient with respect to depth.

Specifically, when considering endeavors like geothermal energy harnessing or hydrogen storage, wellbores are subjected to significant fluctuations in downhole pressure and temperature, leading to dynamic stresses (Zeng et al., 2022). These conditions can challenge the integrity of well seals and may expose vulnerabilities in older wells (Alkhamis and Imqam, 2021). Furthermore, this transition introduces a variety of new fluids, such as geothermal brines and captured carbon dioxide, raising concerns about their chemical compatibility with existing well materials. Such interactions could result in corrosion, structural compromises, or even alterations to reservoir rock permeability (Ndulue et al., 2023).

The integrity and resilience of wells hinge upon their construction materials and methods. For instance, legacy wells often utilize carbon steel casings susceptible to corrosion (Marušić et al., 2006). Innovative treatments like the Tenifer® process have been explored to bolster the corrosion resistance of these casings (Marušić et al., 2006). Furthermore, concerns about the durability of cement sheaths have emerged, especially given the critical role they play in gas migration prevention and ensuring zonal isolation (McDaniel et al., 2014). Fortunately, advancements in cementing techniques, such as the integration of nanomaterials, promise enhanced long-term isolation for oil wells (Jafariesfad et al., 2017).

Beyond materials, the role of advanced diagnostic tools cannot be understated. These tools, ranging from genetic mechanisms to low resistivity pay zone identification, provide vital insights into well conditions and degradation areas (Zhao et al., 2022). They are crucial in ensuring that repurposed wells meet contemporary safety and performance benchmarks. Additionally, with the introduction of new energy applications, the importance of impeccably isolating annular spaces has been underscored (Shinyakov et al., 2018). Legacy wells, due to their historical interventions and possible structural modifications, may face challenges in ensuring such isolation, emphasizing the need for rigorous evaluations and potential reinforcements (Dong and Chen, 2017).

The evolving demands on wells also raise the economic implications

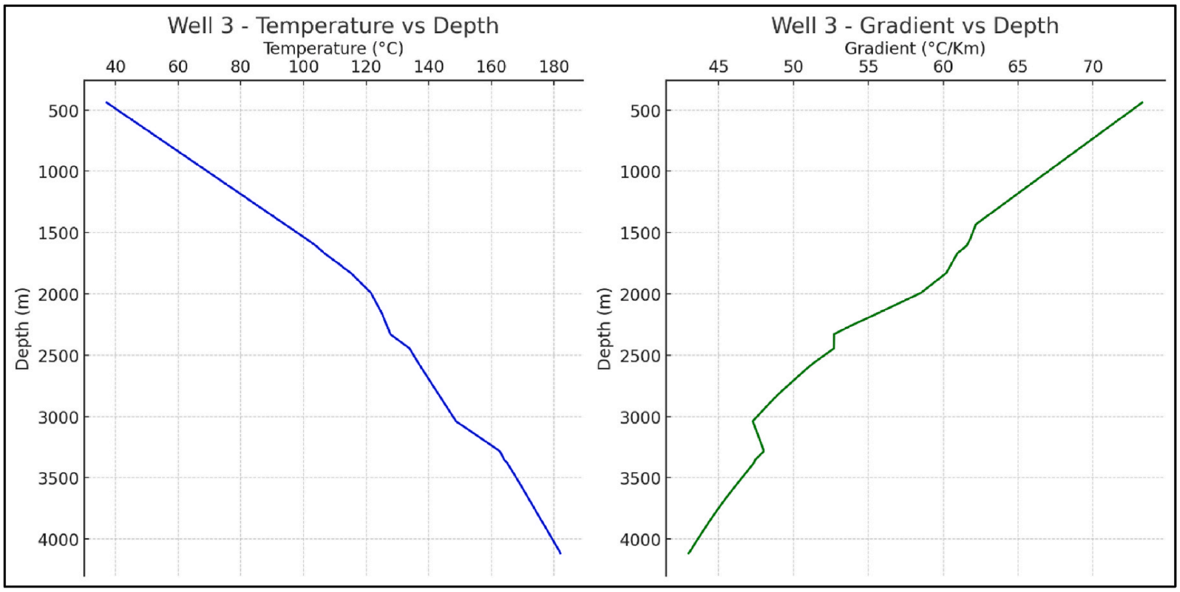

Fig. 7. Behavior of Temperature and gradient with respect to depth.

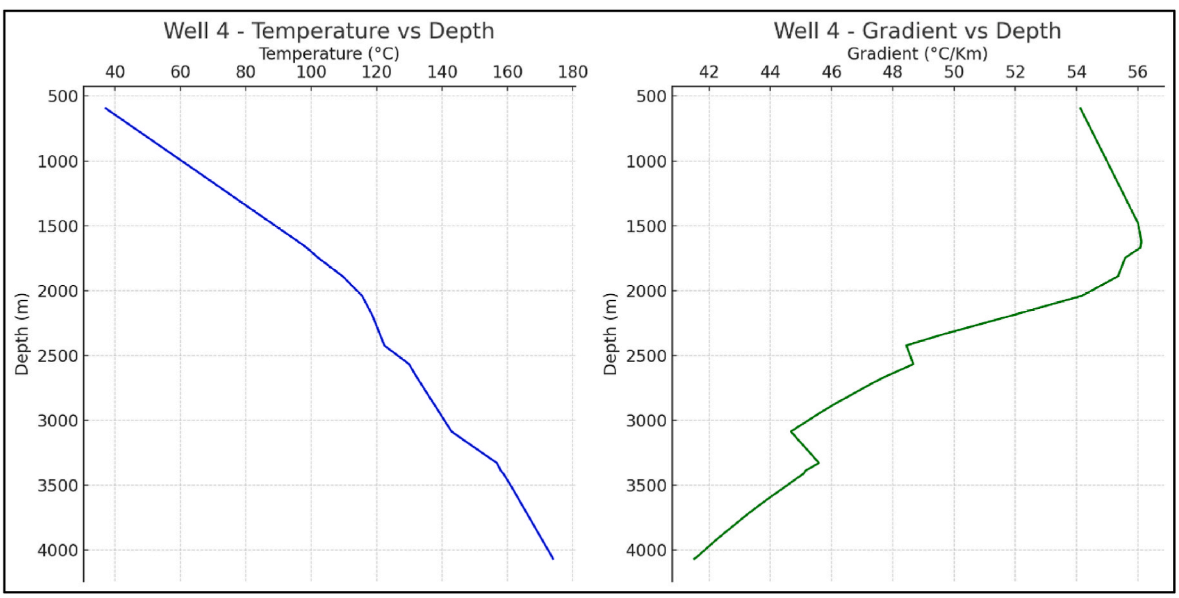

Fig. 8. Behavior of Temperature and gradient with respect to depth.

of retrofitting (Jimenez et al., 2016). While there are direct costs associated with interventions and materials, indirect expenses stemming from potential operational downtimes or environmental consequences also weigh heavily on decisions (Khan et al., 2022). Balancing technical feasibility with economic considerations is pivotal in determining the viability of repurposing a well (Maagi, 2020).

In summary, as the energy sector pivots towards a diverse array of applications, ensuring the longevity, safety, and efficiency of repurposed wells demands a multifaceted approach. This includes advancements in materials and techniques, the implementation of cutting-edge diagnostic tools, and a thorough economic evaluation. Only through such holistic measures can the energy industry confidently and responsibly transition into its next chapter.

3.2. Economic challenges

Repurposing existing oil and gas wells for alternative energy applications holds promise for sustainability goals but brings with it a web of economic challenges. The allure of cost savings through pre-drilled wells is often offset by significant financial outlays required for retrofitting, as outlined by Sathaye et al. (2011). Each alternative energy system, whether for hydrogen storage or carbon capture, necessitates its own set of infrastructure modifications (Tashie-Lewis and Nnabuife, 2021). These adjustments are not merely technological integrations but also repairs for wear and tear inflicted by prolonged hydrocarbon extraction (Tashie-Lewis and Nnabuife, 2021). Compounding these challenges, the overhaul of older wells may entail foundational enhancements, which can be as costly as new installations (United States Environmental Protection Agency, 2021). Unpredictability is another cost factor; undocumented changes or unexpected reservoir conditions can result in unforeseen expenses (Sathaye et al., 2011). Rapid advancements in energy technology further complicate financial planning, as continuous upgrades become necessary (iea, 2020). It is thus vital to conduct comprehensive economic assessments to prevent perceived savings from turning into escalating costs.

Operational expenses also accompany the retrofitting of oil and gas wells for new energy purposes. The complexities of novel energy systems, such as those for geothermal or hydrogen storage, impose continuous and often costly maintenance demands (Ahmad et al., 2022). Aging infrastructures amplify these costs, as ongoing maintenance is critical to staving off the failures that time-induced degradation can bring about. Additionally, adopting the latest technologies and methods, crucial for success, incurs substantial expense, placing financial strains on these projects. The economic feasibility of repurposing projects is also at the mercy of market volatility, a factor that can dictate the outcome of such ventures (Chang et al., 2018). For example, shifts in the hydrogen market's growth or geothermal technology's cost-effectiveness can alter the financial landscape dramatically. Stakeholders, therefore, need to weigh the risks of market fluctuations in their strategic and financial planning (Chuliá et al., 2019). Understanding current trends is crucial, but anticipating future market shifts is equally critical. The energy market's capricious nature means that the viability of options like hydrogen storage or geothermal conversion can swiftly decline due to factors such as oversupply, technological disruptions, or policy changes (Chuliá et al., 2019 & Bade et al., 2023a).

Furthermore, considering the opportunity costs is essential in the economic analysis of well repurposing. Investment in the transformation of an old well has alternative uses, such as funding other energy projects or research into long-term solutions (Issa and Zaid, 2021). This choice requires careful consideration to ensure efficient resource allocation in a rapidly changing global energy landscape. When evaluating investment alternatives, multiple-criteria decision analysis (MCDA) considers numerous factors, from environmental adaptation to installation ease and cost-effectiveness (Dinçer et al., 2022). Renewable energy investments can be analyzed using fuzzy hybrid models, balancing customer expectations with policy considerations (Wang et al., 2019). These methods help to evaluate investment choices holistically.

Economic and environmental sustainability must also guide repurposing decisions. The shift to sustainable energy needs a careful path, which includes policy implications (D'Alessandro et al., 2010). Techno-economic analysis (TEA) quantifies the feasibility of converting manufacturing facilities to produce sustainable aviation fuel, considering both economic viability and ecological impact (Brandt et al., 2022). This analytical approach informs the potential economic and environmental benefits of repurposing initiatives.

3.2.1. *Cost of conversion*

Converting outdated oil and gas wells to geothermal energy producers involves significant expenditure. Costs can vary widely, from $2 to $7 million per well, influenced by several factors, including location and the well's depth, as reported by Watson et al. (2020). The same study scrutinized onshore fields in the UK to determine their suitability for transforming hydrocarbon wells to geothermal applications, using aggregated water production data from operating wells to evaluate their geothermal prospects. Carbon capture and storage (CCS) projects also come with a notable price tag. Mwelu et al. (2018) suggest that capturing CO_2 can cost between $50 and $100 per metric ton, values that fluctuate based on the technology used and the operational scale. Retrofitting existing wells to accommodate CCS would also involve supplementary expenses for transportation and injection infrastructure, potentially adjusting the aforementioned cost estimates. Data specific to the costs of modifying oil and gas wells for hydrogen storage purposes is less precise. However, insights from the U.S. Department of Energy suggest that storing hydrogen in salt caverns typically incurs costs of approximately $5 to $12 per kilogram of hydrogen (Mwelu et al., 2018). Adapting these figures for the conversion of existing wells would necessitate additional research and analysis.

Furthermore, regulatory compliance poses another cost layer in various regions. For example, in certain U.S. locales, the expense of sealing an abandoned oil or gas well can span from $20,000 to $145,000, determined by the well's depth and its location (Mwelu et al., 2018). Repurposing these wells for renewable energy ventures might provide a way to circumvent these abandonment costs, although adherence to regulatory standards is a critical factor in the planning and execution of such projects. The presence of government tax incentives and subsidies can also play a pivotal role in determining the viability of well conversion projects. In the U.S., the 45Q tax credit offers a substantial benefit of $50 per metric ton of CO_2 sequestered, thus potentially boosting the adoption of CCS techniques. Busom et al. (2014) discuss the broader implications of such fiscal measures, noting their influence on encouraging business research and development. These financial incentives are crucial in fostering the transition of old wells to alternative, more sustainable energy uses.

3.3. *Social and environmental challenges*

The transition of oil and gas wells to alternative energy platforms entails a series of social considerations. Central to this is the matter of stakeholder engagement, as Melbourne-Thomas et al. (2021) illustrate through the "rigs to reefs" initiative in Australia, wherein the community's role in determining the use of decommissioned rigs as recreational fishing sites was paramount. This case underscores the importance of inclusive stakeholder involvement to achieve successful and socially accepted repurposing projects. In addition, the shift away from fossil fuel exploitation poses distinct socio-economic challenges, particularly for oil-dependent developing nations. Adams et al. (2019) explore the "resource curse," detailing how mismanaged oil revenues and lack of corporate social responsibility (CSR) compliance can impede socio-economic progress. Addressing these issues is vital to ensure that well repurposing aligns with the goals of sustainable development and supports the broader social fabric of these regions. On the environmental front, repurposing offers solutions to the hazards posed by orphaned wells, which Kang et al. (2021) note are prone to leaking harmful substances. Thus, converting these wells not only mitigates environmental risks but also presents an opportunity to realize economic and ecological gains.

Yet environmental challenges persist, as indicated by the research of Bu et al. (2012) on the utilization of abandoned wells for geothermal energy, which may alter the geothermal gradient and fluid chemistry. Such changes necessitate a thorough understanding and effective management to ensure environmental integrity during the repurposing process. Moreover, the rigorous evaluation of Environmental Impact Statements (EIS) is pivotal in identifying and managing potential environmental repercussions, as Anifowose et al. (2016) demonstrate in their quality assessment of EIS within the oil and gas sector. They point out that environmental concerns such as emissions, odors, noise pollution, and soil degradation must be meticulously assessed and mitigated to guarantee that the transformation of oil and gas wells contributes positively to environmental sustainability.

In conclusion, repurposing oil and gas wells for alternative energy applications represents a promising avenue for progress toward a more sustainable and environmentally friendly energy future. To realize this potential, industry leaders, policymakers, and communities must work in unison, guided by rigorous scientific research and a clear-eyed appraisal of the economic realities. The journey will not be without its hurdles, but with a dedicated and holistic approach, these challenges can be transformed into the bedrock for a new era of energy production that honors our environmental responsibilities and harnesses the power of innovation for the greater good.

4. Different uses of abandoned oil wells

There are numerous uses for abandoned oil wells, which greatly increase the versatility and efficiency of energy use. One primary application of geothermal energy will be in electricity generation, where it can offer a reliable and constant power supply, unlike intermittent renewable sources such as wind and solar. Geothermal energy can also provide heating for residential, commercial, or industrial entities, serving as a sustainable alternative. Another crucial applicationinvolves storing the extracted thermal energy in various forms, such as hot water reservoirs, for future use. In this context, we can utilize stored energy to manage supply and demand, particularly during periods of high energy demand or low supply from renewable sources. Desalination processes can also utilize geothermal energy, as the generated heat converts seawater into fresh drinking water. Utilizing geothermal energy from abandoned wells offers a multi-pronged approach to increasing energy sustainability and serving various energy needs.

4.1. Power generation

Abandoned wells can be retrofitted for geothermal energy utilization and power production. The wells selected must have petroleum recoveries that fall below the economic index or limit Initially, before integrating the power cycles, which can be dry steam, flash steam, or binary cycle plants, into the abandoned wells, there is an estimation of the reservoir properties such as thermal energy, water density, pressure, and recoverable factor. The next step is to use a computer model that includes heat transfers, energy exchange equations, and the extraction well for geothermal reservoirs that are mostly made up of wet steam. Finally, the power plants are selected based on the temperature output from the extraction well, which varies between high, medium, and low-temperature ranges (Izula et al., 2022; Rafferty et al., 2000). Typically, binary plants are efficient for direct power generation from abandoned wells and are also used for low-temperature applications (Alimonti and Soldo, 2016).

A binary-cycle power-generating system is a type of geothermal power plant that transforms thermal energy from the rock into electricity using a two-step heat exchange mechanism. The first process involves passing a working fluid through a double-pipe exchanger, where the rock heats it. The Organic Rankine Cycle (ORC) device's second stage involves the exchange of heat between the working fluid and a fluid with a low boiling point (Cortez et al., 1973). The first step of the process often employs a high-boiling-point liquid, such as isobutane or pentane, as the working fluid. The hot rock circulates this fluid through a closed loop in the geothermal reservoir, heating it. After entering the double-pipe exchanger, the hot working fluid transfers its heat to a different fluid loop that travels through the ORC machine. A fluid with a low boiling point, like a refrigerant, serves as the ORC machine's operating fluid. The heated working fluid's heat evaporates, and the resulting steam powers a turbine that generates energy. After the low-boiling-point fluid evaporates and condenses back into a liquid, we repeat the procedure by exchanging heat with a cooling fluid, usually water (Cortez et al., 1973).

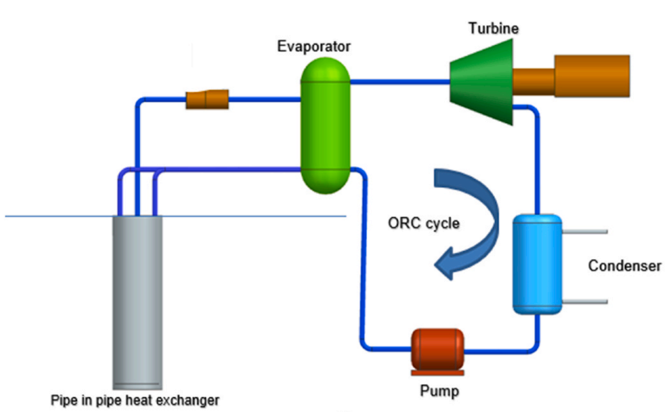

Fig. 9. Power generation using orc from a vertical well as a double pipe heat exchanger (Closed loop system).

Overall, generating electricity from geothermal energy using a binary-cycle power generation system is effective and environmentally friendly. Fig. 9 illustrates the schematic diagram of a pipe in pipe configuration closed loop geothermal system with a refrigerant operating as a secondary or working fluid to produce electricity. It has minimal negative effects on the environment and doesn't emit any greenhouse gases or other pollutants. Moreover, it is a dependable power source that can run continuously, seven days a week, making it a desirable alternative for areas with significant geothermal potential. Since this is still a relatively new field of study, there isn't much information available about the electricity produced from abandoned wells in the USA at the moment. Yet, according to estimates from the US Department of Energy, the overall energy potential from defunct oil and gas wells might be up to 100 GW of electricity-producing capacity. Generally, abandoned wells provide a potential geothermal energy source that could help reduce reliance on fossil fuels and greenhouse gas emissions, despite ongoing research and development in this field.

Wells in Qatar are on the verge of closure due to temperatures exceeding 149 °C in active wells. Therefore, the current work examines a binary geothermal power production system from thermodynamic and economic perspectives for producing commercial electricity. Kharseh et al. (2015) found that the plant produces 5 MW of power at a levelized cost of 5.2–5.6 kWh. In addition to the ORC, oil wells also generate electricity using thermoelectric power generators (TEGs). Wang et al. (2018a,b) developed a prototype design and an analytical model to calculate the power generation efficiency using TEGs.

Rafferty et al. (2000) demonstrated a case study to generate a 100-kW plant with a low-temperature availability in the subsurface of around 200 °F. The case study uses a flow rate of 18.92 l/s, which provides a plant efficiency of close to 8.5%. Alimonti and Soldo (2016) asserts that a low-temperature input of approximately 200 °F and a well depth of approximately 10,000 feet can generate a thermal efficiency of approximately 8%, resulting in a power output of 900 kW. In this case, the mass flow rate is close to 5.5 l/s, which is on the lower side. These studies prove the case that it is possible to produce power from abandoned oil wells, irrespective of the low temperature and mass flow rate availability. The above case study demonstrates that integration on power cycles like ORC is best suited to the sub surface's varying physical and chemical properties.

4.2. District heating

Ground-source heat pumps (GSHP) are among the most important geothermal applications. Applications for space heating and cooling have utilized GSHP systems. Ground-source heat pumps use the earth's constant temperature to heat and cool buildings. Ground-source heat pump systems, potentially more efficient than typical air-source heat pumps, can utilize abandoned oil wells as boreholes. Abandoned wells with greater depths contain abundant geothermal energy for heating and cooling applications (Nian et al., 2019). Many oil and gas wells have either completed their production or ceased operations. Implementing the coaxial heat exchanger technology, typically employed in shallow geothermal energy extraction, could potentially exploit these wells for geothermal energy generation.

According to Caulk and Tomac (2017), abandoned wells in the counties of Santa Clara, Monterey, and Santa Barbara in California may be suited for direct-use low-temperature enhanced geothermal systems (EGS), such as district heating or greenhouse heating. At depths of 900–2000 m, low-to moderately consolidated sandstone rock layers enclosed by shale layers with similar consolidation characteristics provide a viable reservoir for geothermal energy extraction. Temperatures ranging from 40 to 70 °C, with some wells reaching up to 90 °C, are also appropriate for direct-use, low-temperature EGS applications like district heating or greenhouse heating.

The majority of direct-use applications for geothermal fluids operate at temperatures ranging from 50 °C to 150 °C. These temperatures are appropriate for a wide range of uses, including room heating, greenhouse heating, and industrial processes (Lund, 1997). Most documented geothermal systems in the United States, for example, have temperatures around 90 °C, with only 5% having temperatures above 150 °C. As a result, low-to moderate-temperature geothermal systems are especially significant for direct-use applications in the United States and other countries where they are common (Lund, 1997; Smith and Shaw, 1979).

From the case study of Macenić and Kurevija (2018), the deep well Pčelić located in the Drava sub-basin had its energy and temperature variations of the fluid calculated over a 20-year operational period for two scenarios—one with a constant heat load throughout the year and the other with a variable heat load that relies on the temperature of the outside air. Over 20 years, direct industrial heating could achieve a maximum theoretical heat extraction rate of 400 kW, with the fluid temperature stabilizing at an ideal 50 °C. The primary goal of the project in Mandaree, North Dakota, is to directly use hot water from the Inyan Kara formation. The system arrangement consists of two abandoned wells, set up in a doublet configuration with one injector and one producer. The system layout places the two wells 1400 m from the town and separates them by 1250 m. Both wells are vertical and feature existing 7-inch perforated casings that enable geothermal fluid to flow at a rate of about 40–50 l/s. Once the in-situ heat requirement in Mandaree, which is close to 4.1 MW thermal, has been satisfied, the well output can supply an additional 1 MW thermal for operating commercial greenhouse activities(Eagle-Bluestone et al., 2022).

The first study that looked at how coproduced water from wells in the Italian oil field of Villafortuna-Trecate could be used to generate geothermal energy found that the coproduced hot water could be used for a number of things, such as direct district heating, using an Organic Rankine Cycle (ORC) plant to make electricity, and co-generating heat and power. The findings indicate that selecting the power generation option could potentially generate 500 kW of installed electric power, or around 25 GWh, from a single well that empties a cylindrical reservoir over 10 years. The cogeneration option, on the other hand, appears more enticing because it can produce a lower net electrical output of 143 kW per well while simultaneously serving about 46 utilities per well with a 660-kW district heating plant (Alimonti et al., 2014). Fig. 10 illustrates the schematic diagram of cogenerating heat and power geothermal plant where the recovered heat from the Rankine cycle is used for district heating.

Caulk and Tomac (2017) and Alimonti et al. (2014) demonstrated that it is possible to implement both EGS and GSHP using abandoned oil wells in California with a temperature availability of around 260 F. In this case, the mass flow rate ranges from 1 to 10 l/s. Lund (1997) creates a similar case with the same mass flow rate, but the temperature is considered to be between 130 and 240 °F. Eagle-Bluestone et al., 2022 and Alimonti et al. (2014) demonstrated the feasibility of operating a cogeneration plant up to 1 MW, provided the thermal capacity, temperature, and mass flow rate are low. All of the above case studies is summarized in Table 3 to demonstrate that local district heating is an efficient method that can adapt to the varying physical conditions of the reservoir using abandoned oil and gas wells. It is suitable for use in both high- and low-temperature conditions, as well as mass flow rate conditions.

4.3. Energy storage

Energy storage in decommissioned oil wells entails using these wells to store a variety of forms of energy, including thermal, pumped hydro, and compressed air. The idea is to utilize the wells' subsurface reservoirs to store energy during times of excess supply and release it during times of high demand (Matos et al., 2019). This strategy offers several benefits, such as using existing infrastructure and avoiding the need to build new energy storage facilities, which can be costly and have a greater environmental impact. Additionally, in areas with favorable geological conditions, abandoned oil wells can provide a practical solution for energy storage.

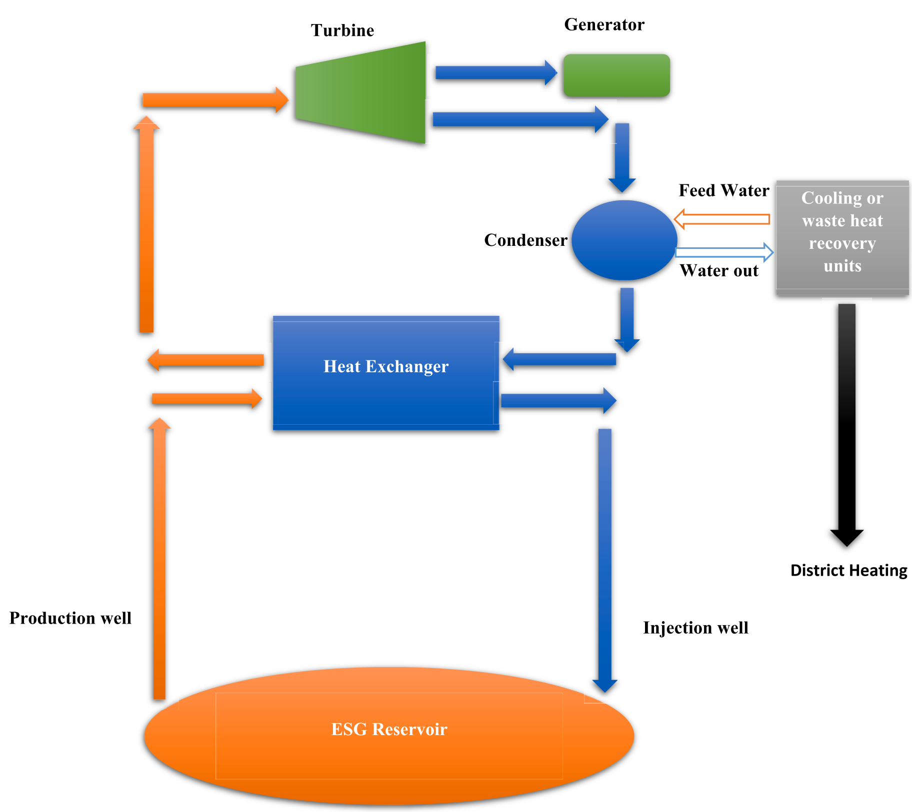

Fig. 10. Combined power and heat binary geothermal plant (modified from, Huddlestone-Holmes and Hayward, 2011).

Table 3
Case studies and findings for using Abandoned oil wells for district heating/cooling.

Case studies	Findings
Characteristics of abandoned wells in California (Caulk and Tomac, 2017).	Temperatures ranging from 90 ∘c, are also appropriate for direct use low-temperature EGS applications like district heating or greenhouse heating.
Direct utilization of Geothermal resources (Lund, 1997).	Geothermal fluids operate at temperatures ranging from 50 °C to 150 °C, these temperatures are appropriate for a wide range of uses, including heating.
Study on deep well Pčelić located in the Drava subbasin (Macenić and Kurevija, 2018).	A maximum theoretical heat extraction rate of 400 kW could be attained for direct industrial heating, with the fluid temperature stabilizing at an ideal temperature of 50 °C.
Case study on abandoned wells in Inynankara formation in Mandree ND (Bluestone et al., 2022).	The well output can supply an additional 1 Mw Thermal, which can be used for operating commercial greenhouse activities, after the in-situ heat requirement in Mandaree.
Assessment of the potential for using coproduced water from wells in the Italian oil field (Alimonti et al., 2014).	The Cogeneration plant can produce lower net electrical output of 143 kW per well while at the same time, a 660-kW district heating plant.

Krishna et al., 2022 studied large-scale subsurface energy storage in saline aquifers using idle oil and gas wells and found that converting idle wells into energy storage wells can be advantageous. It provides a helpful option for power utilities while also resulting in a favorable financial outcome for oil and gas businesses. Operators can convert these underutilized wells into access routes to saline aquifers for storing compressed air. The design of surface facilities, along with modelling and calculations of subsurface conditions, demonstrates the economic viability of the suggested method for large-scale energy storage.

We currently use Underground Thermal Energy Storage (UTES) to utilize industrial waste heat. Depleted oil wells typically carry out this process (Xie et al., 2018). With this technology, it is possible to repurpose existing infrastructure and reduce the site's environmental impact by retrofitting abandoned oil wells. Overall, the installation of a coaxial borehole heat exchanger with insulation could be a creative way to reuse abandoned oil wells and provide nearby areas with access to seasonal thermal energy storage (TES) (Xie et al., 2018; Matos et al., 2019). Xie et al. (2018) concluded that without the need for costly drilling, groundwater extraction, and recharge, it is possible to turn a depleted oil well into an underground thermal energy storage (UTES) system for seasonal heat extraction and storage. Also, this method prevents problems with groundwater recession, corrosion, and scaling. Drilling the well to the right depth will maximize storage efficiency if a heat source with a certain temperature already exists. On the other hand, the minimum inflow temperature necessary to maintain heat storage must be considered if an existing well has a certain depth. Quin et al. (2021) have developed a study on compressed wind energy storage using abandoned wells, which not only eliminates the need for storage vessels but also facilitates the implementation of an isothermal process for compressed air storage, resulting in improved round-trip efficiency. A case study estimates that a typical wind farm can reduce the cost of valuable energy by over 10%. Additionally, the simulated dispatchability ratio has a potential range of 86.7%, significantly surpassing the 55.7% level that a wind farm without energy storage achieves. Moreover, carbon capture and storage (CCS) projects, which involve capturing and storing carbon dioxide emissions from factories, power stations, and other industrial facilities underground, can utilize abandoned wells. Raza et al. (2017) concluded that several well parameters, such as cement-sheath integrity, distance from faults and fractures, and well depth, are crucial for the efficient storage of CO. In the context of hydrogen storage, factors such as the integrity of the wells, the potential for leaks, and other environmental factors all play a significant role. Aside from that, it could be expensive to maintain and convert abandoned wells for energy storage (Ugrate et al., 2022). Much research has concluded that storage of hydrogen in abandoned wells is possible, but many factors that lead to the loss of well integrity must be considered for H2, CH_4, and CO_2 storage (Bai et al., 2014; Ugarte and Salehi, 2021; Bade et al., 2023; Bade et al., 2023a). Table 4 represents summation of case studies and findings related to abandoned wells for energy storage. Overall, while the conversion of abandoned wells for energy storage entails certain technical and economic challenges, the potential benefits in terms of sustainability, cost savings, and environmental impact make it a worthwhile pursuit. Continued research and technological advancements will further enhance the feasibility and efficiency of these energy storage methods, contributing to a more resilient and sustainable energy future.

4.4. Desalination using abandoned oil and gas wells

An efficient way to produce electricity and clean water while reusing existing infrastructure is to build a system that combines power generation and water purification using defunct oil wells. To guarantee the project's safety and viability, careful planning and execution are required. A new approach studied by Kolahi et al. (2022) involves installing a combined power generation and water purification system. The system comprises two ORC cascades, upper and lower. The evaporator serves as the condenser in the upper ORC cascade, while the lower ORC employs zeotropic mixtures, with the Humidification Dehumidification Desalination (HDD) units connected to the evaporator. The air circulating between the humidifier and dehumidifier cools and condenses as salty water passes through the dehumidifier. The heated saline water in the heater then warms and humidifies the airflow into the humidifier. By repeating this procedure, the dehumidifier eventually produces freshwater, releasing the residual water as brine (Kolahi et al., 2020; 2022).

The findings revealed that the total output power was between 22 and 23 kW, and the amount of freshwater produced per day was between 532,750 and 533,042 l. Norouzi et al. (2022) conducted research on the utilization of abandoned oil and gas wells for a geothermal seawater desalination system (GSWDS) in the Ahwaz oil field in southern Iran. This approach aims to decrease emissions produced by conventional desalination systems, which rely on burning fossil fuels and produce

Table 4
Case studies and findings for using Abandoned oil wells for Energy storage.

Case studies	Findings
Subsurface reservoirs store energy during times of excess supply (Maltos et al., 2018).	In areas with good geological conditions, abandoned oil wells can offer a practical answer for energy storage.
Large-Scale Subsurface Energy Storage in Saline Aquifers (Krishna et al., 2022).	Converting idle wells into energy storage wells can be advantageous and provide a helpful option for power utilities and also favorable financial outcomes for oil and gas businesses.
Underground Thermal energy storage using abandoned oil wells as Coaxial Borehole heat exchangers (Xie et al., 2018).	A coaxial borehole heat exchanger with insulation could be a creative way to reuse abandoned oil wells and give the nearby areas access to seasonal Thermal energy storage.
Compressed wind energy storage using abandoned wells (Quin and Loth, 2021).	The drop in cost and the dispatchability ratio is higher for the farms with energy storage using abandoned oil wells.
Guideline to be followed about sophisticated wells for storage and injection (Raza et al., 2017).	Well parameters like cement-sheath integrity, distance from faults and fractures, and the depth of wells are important for efficient storage of Co2 without leakage.
Hydrogen storage possibilities and challenges (Bai et al., 2014; Ugrate et al., 2022).	Storage of Hydrogen in abandoned wells is possible but many factors which lead to the loss of well integrity must be considered.

harmful emissions like oxides of nitrogen (NOx), carbon monoxide (CO), and sulfur dioxide (SO2) (Younos, 2005). Several researchers have created a conceptual design of a multistage desalination system that uses waste heat, and it is the base idea to use abandoned wells as a heat source for the desalination process (Abdel-Jabbar et al., 2007; Ameri and Eshaghi, 2016; El-Dessouky et al., 2000). In GSWDS, water, reaching nearly 100 °C, transfers heat from the abandoned wells to the desalination units. This heat is transferred to the heat exchanger in the desalination units. The findings of this study demonstrated that, in comparison to conventional desalination systems, a single abandoned oil well may produce 565 m^3 of fresh water from seawater and significantly lower emission generation.

Kiaghadi et al. (2017) modelled the geothermal energy efficiency of abandoned oil and gas wells to desalinate generated water illustrated in Fig. 11, where the desalination units absorb the heat from the heated fluid stream. The extracted hot water powers the desalination units. The study also demonstrated that a 4000-m-deep well with a geothermal gradient of 0.05 °C/m can effectively purify water with up to 170,000 mg/l of total dissolved solids. Despite its high concentration, the well can still generate about 600,000 l of pure water per day.

Tyzer (2020) conducted a study on abandoned oil wells for produced water desalination and found that many hot water-enriched abandoned wells in the Polish lowlands and Podhale are suitable for combined desalination and geothermal potential. Norouzi et al. (2022) demonstrated that multistage desalination with secondary preheating can also achieve desalination in abandoned wells within a low to medium temperature range.

4.5. Overall observations of different uses of abandoned oil wells

Among the various uses of abandoned oil wells, power generation, district heating, and cooling are particularly well-suited due to their adaptability to varying physical conditions. Geothermal power generation, especially with binary cycle plants, can efficiently utilize a broad range of temperatures and mass flow rates, allowing customization to the specific thermal and flow characteristics of an abandoned oil well. Binary cycle plants, for instance, can operate with resource temperatures as low as 220 °F, making them versatile for different geothermal conditions. This flexibility enables the conversion of geothermal energy into electricity, even in the presence of poor thermal properties.

District heating and cooling systems similarly benefit from this flexibility, as they can effectively operate with diverse geothermal temperatures and varying flow rates. These systems use geothermal heat directly for heating buildings or for cooling purposes through absorption chillers. They are particularly efficient because they bypass the need for high temperatures, which are required for power generation, and instead utilize moderate to low temperatures efficiently.

Additionally, both applications leverage the existing infrastructure of abandoned oil wells, reducing the need for extensive new drilling and associated environmental impacts. This makes them cost-effective and environmentally friendly solutions. These systems adapt to different subsurface conditions, enabling deployment in a variety of geological settings, maximizing the utilization of available geothermal resources, and contributing to sustainable energy solutions.

In contrast, energy storage in saline aquifers, hydrogen or CO2 storage, and desalination using abandoned wells do not offer the same diversity in operating conditions. These applications are more heavily reliant on the subsurface's specific physical properties and chemical composition. This dependence can limit their effectiveness and adaptability compared to power generation and district heating and cooling. Furthermore, these applications are often more expensive due to the need for extensive infrastructure modifications and the complexity of the processes involved.

5. Discussions

The increased number of abandoned oil wells in the United States poses both obstacles and opportunities, notably in terms of potential repurposing for geothermal energy production. Geothermal energy extraction from abandoned oil wells represents an exceptional opportunity in the renewable energy sector. The basic assumption is that the Earth's natural geothermal gradient, which is a temperature rise with depth that is especially steep in volcanic locations, is used. Converting abandoned oil wells to geothermal energy sources is not only a novel renewable energy approach but also a way to address the growing number of orphaned wells, particularly in the United States. These abandoned wells frequently still have high temperatures and temperature gradients, making them attractive candidates for geothermal applications.

The Red River Formation in North Dakota's Williston Basin has enough temperature and temperature gradient to extract geothermal energy, according to a case study using scout ticket data from the NDGS. However, this repurposing is not without complications, including mechanical, social, and environmental issues, as well as the exploration of

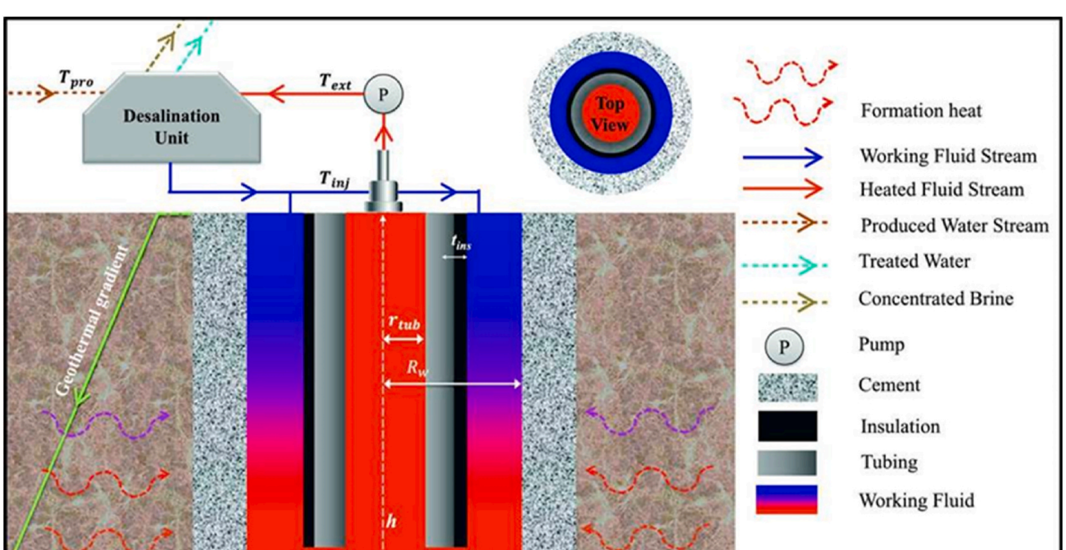

Fig. 11. Schematic representation of Geothermal heat extraction which uses heat to treat the produced water on the surface (Kiaghadi et al., 2017; Wight and Bennett, 2015).

alternate uses for these wells.

Aside from geothermal energy production, abandoned oil wells have other applications. Converting them for underground natural gas, carbon dioxide, or hydrogen storage can aid in energy storage and carbon capture operations. Other potential uses include water desalination, district heating, and finally electricity production using a binary cycle for low-temperature applications and a flash cycle for high-temperature availability.

The challenges of repurposing abandoned oil wells include ensuring structural integrity because older wells, constructed with then-available technology, lack the durability of current engineering and must contend with issues such as changing pressures, chemical incompatibilities, and structural vulnerabilities. Adapting existing wells for new uses necessitates the use of novel materials, sophisticated diagnostics, and careful economic analysis. Factors like location, depth, and technology selection influence the costs of these adaptations. Regulatory compliance and government incentives also influence the financial sustainability of such conversions. Transitioning oil and gas wells to alternate uses necessitates broad stakeholder participation and the resolution of socioeconomic issues, particularly in oil-dependent countries. Environmental factors, such as reducing dangers from orphaned wells and managing variations in geothermal gradients, are critical for long-term reuse.

6. Conclusions

- All oil-producing states possess the requisite physical and chemical conditions for the implementation of geothermal energy, owing to the elevated temperatures found in oil reservoirs.
- Converting oil and gas wells into geothermal energy sources requires addressing technological, economic, social, and environmental obstacles.
- To accomplish successful transformation, a comprehensive approach is required, which includes the use of advanced technology, active involvement of the community, responsible management of the environment, and support from legislation. This will enable the conversion of decommissioned resources into valuable assets that contribute to a sustainable energy future.
- The rise of abandoned oil wells in the United States offers substantial prospects for renewable energy generation and other advantageous uses, such as heating, storage, and desalination.

CRediT authorship contribution statement

Ajan Meenakshisundaram: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Olusegun Stanley Tomomewo:** Writing – original draft, Conceptualization. **Laalam Aimen:** Writing – original draft, Software, Conceptualization. **Shree Om Bade:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Nomenclatures

CCUS	Carbon Capture Utilization and Storage
EPA	Environmental Protection Agency
NDGS	North Dakota Geological Survey
OG	Oil and gas
AB	Abandoned
MCDA	Multiple Criteria Decision Analysis
TEA	Technoeconomic Analysis
CCS	Carbon Capture and Storage
CSR	Corporate Social Responsibility
EIS	Environmental Impact Statements
ORC	Organic Rankine Cycle
TEG	Thermo Electric Generator
GSHP	Ground Source Heat Pumps
EGS	Enhanced Geothermal Systems
UTES	Underground Thermal Energy Storage
TES	Thermal Energy Storage
HDD	Humidification Dehumidification Desalination
GSWDS	Geothermal Sea water Desalination Systems
CO	Carbon Monoxide
CO2	Carbon Dioxide
NOx	Oxides of Nitrogen
SO2	Sulfur Dioxide
H2	Hydrogen
CH4	Methane
w/m2	Watts per meter square
$^{\circ}c$	*Degree Celsius*
$^{\circ}F$	*Degree Fahrenheit*
m	Meter
m2	Meter Square
W/m/K	Watt per meter per Kelvin
Kw	Kilo watt
GWh	Giga watt hour
KWh	Kilowatt Thermal
l/s	Liter per second
Kj/kg	Kilojoule per kilogram
Kg/m2	Kilogram per cubic meter
m/s	Meter per second
MJ	Megajoule
MW	Mega watt
mW/m2	Milliwatts per square meter

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